MDA692 Data Analytics Capstone Project

Assignment 2: Project Implementation and Evaluation Report

Hotspots of Cybercrime in Identifying High-Risk Areas through Geospatial Analysis

Group 2

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Item	Description
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Abstract

The project seeks to solve the practical difficulty of turning cyber-attack event records into geospatial intelligence so that cyber analysts, security teams, and decision makers might more easily identify high-risk areas. The adopted approach applies spatial analysis, clustering, supervised learning, model evaluation and explainability methods as well as Python-based data processing, IP geolocation, and contextual data enrichment. Starting with cybersecurity event logs, the project adds coordinates, country, city, autonomous system, internet use, GDP per capita, population density, and raster-based local population density to make them better. It then trains classification models to predict high-risk records, generates engineered risk variables, computes a final hotspot score, and assesses clustering patterns.

Random Forest has the highest single model in the recorded implementation with an accuracy of 0.8524. Also considered were XGBoost, LightGBM, CatBoost, and an ensemble model. While the spatial component is supported by Getis-Ord Gi* hotspot mapping, kernel density estimation, a correlation heatmap, network graph analysis, and SHAP explainability, a confusion matrix backs the last prediction output. Moran's I is also reported as 0.00339 in the implementation. This shows that there is very little positive spatial autocorrelation in the anomaly-score layer. So, the results should be looked at very carefully. The project presents a working analytical prototype rather than a full-fledged production system. Its main contribution is the combination of cyber-attack data, IP-based geolocation, contextual enrichment, and explainable machine learning into a clear geospatial cybercrime risk assessment process.

1. Introduction

Individuals, businesses, governments, and digital infrastructure are continually under attack by cybercrime. Though many cyber events happen through virtual networks, the information linked to them usually includes temporal and geographical indications. From source IP addresses, network ownership, country-level indicators, timestamps and attack forms can be created spatial and analytical characteristics. These elements, when examined collectively, can help security staff give priority to areas with high risk and assist in spotting likely cybercrime hotspots.

As an explanatory and predictive research problem, the project theme Hotspots of Cybercrime in Identifying High-Risk Areas through Geospatial Analysis is meant to be The project is driven by the

necessity of knowing how useful geographic, statistical, and predictive insights may be derived from cyber-attack records. The report emphasizes the applied analytical approach, which includes dataset loading, data cleaning, cyber-attack record enrichment, feature generation, spatial visualisation, machine learning model evaluation, and results interpretation. The focus is on actually using the suggested solution instead of just thinking about how it could work.

The approach is intentionally based on data. It covers the parts of the completed study that were used and checked, such as data cleaning, geolocation, contextual merging, risk-score construction, DBSCAN clustering, Getis-Ord Gi* mapping, kernel density estimation, correlation analysis, network graphing, classification modelling, SHAP feature explanation, and Power BI dashboard reporting. It makes no claim of completed deployment capabilities such as a PostGIS application layer, live data pipeline, or production cloud service since the completed implementation stage reveals no such parts. For preserving a fair assessment of the project and for academic honesty, this limit is crucial.

1.1 Dataset Description and Sources

One main cybersecurity event dataset is used by the application together with a number of supporting enrichment datasets. The local working file updated_cybersecurity_attacks.csv, which is derived from the openly accessible Kaggle Cyber Security Attacks dataset [10], is the main source of information. Event-level records pertaining to cyber events and network activity abound in the dataset. Variables employed in the implementation include Timestamp, Source IP Address, Destination IP Address, Protocol, Traffic Type, Attack Type, Severity Level, and Anomaly Scores. Timestamps enable temporal analysis, IP addresses enable geolocation and network aggregation, attack categories enable behavioral interpretation, severity levels enable risk labelling, and anomaly scores enable hotspot scoring—all of which provide the technical foundation for the project.

Geospatial analysis requires a cyber log's automatic provision of trustworthy latitude and longitude; hence, the basic dataset alone is insufficient. This is why the implementation makes the source IP addresses better by using MaxMind GeoLite2 data [11]. Derivation of Latitude, Longitude, Country, and City from GeoLite2-City.mmdb and autonomous system number and ASN organisation from GeoLite2-ASN.mmdb depend on These factors turn the event log into a spatial dataset and give network-infrastructure background. The report nonetheless analyzes these values precisely since IP geolocation is not the actual physical position of an attacker or victim but rather may reflect service provider infrastructure, routing location, VPN endpoints, or cloud hosting regions.

Also utilized are three country-level socioeconomic datasets: population_density.csv, gdp_per_capita.csv, and internet_users.csv. These files are aligned with World Bank-style World Development Indicator variables and are reduced in the implementation to Country Name and 2023 [YR2023] values [12], [13]. Before being combined with the cyber-attack data, the variables are changed to Country, Internet_Users, GDP_Per_Capita, and Population_Density. These elements give a general

country framework for understanding if internet access, financial situation, or population distribution correlate with attack concentration. These should not be read as exact local city-level indicators as they are country-level factors.

Through a WorldPop raster file called global_pop_2020_CN_1km_R2025A_UA_v1.tif [14], the implementation also brings in local population information. Opening the raster with rasterio and sampling at event coordinates yields local_population_density. Since it relies on gridded population values close to every mapped event, this factor is more geographically specific than the country-level population-density CSV. These datasets taken together let the study go from a basic cyber-event table to a more complex geospatial risk dataset with technical, geographic, socioeconomic, demographic, and network-based factors.

Table 1: Datasets and enrichment sources used in the implemented analysis.

Dataset / file	Source	Variables used	Role in implementation
updated_cybersecurity_attacks.csv	Kaggle Cyber Security Attacks dataset [10]	Timestamp, source and destination IPs, protocol, traffic type, attack type, severity, anomaly score	Main event-level cybersecurity dataset for cleaning, labelling, modelling and risk scoring.
GeoLite2-City.mmdb	MaxMind GeoLite2 [11]	Latitude, longitude, country, city	Converts source IP addresses into geographic coordinates and place labels.
GeoLite2-ASN.mmdb	MaxMind GeoLite2 [11]	ASN, ASN organisation	Adds network ownership context and supports ASN-level risk features.
internet_users.csv, gdp_per_capita.csv, population_density.csv	World Bank / World Development Indicators [12], [13]	Internet users, GDP per capita, population density	Adds country-level socioeconomic and connectivity context.
global_pop_2020_CN_1km_R2025A_UA_v1.tif	WorldPop [14]	Raster population value sampled at event coordinates	Creates local population density for spatial analysis and modelling.

1.2 Project Scope and Report Structure

This study covers the applied analytical technique and its evaluation. Section 2 specifies the evaluation standards and project goals. Section 3 details the implementation process from data loading via spatial analysis, machine learning, and Power BI visualisation. Testing and problem-solving are the subjects of Section 4. Section 5 examines the outcomes using dashboard outputs, model measures, and spatial

visual proof. Section 6 examines the execution against the results of the MDA691 literature review. Section 7 presents the professional issues and project contributions. The study ends in Section 8 with a discussion of limitations and future directions.

2. Project Objectives and Evaluation Criteria

Four key project goals derived from the general research design and implementation criteria were used to assess the implementation. These goals link the research strategy with concrete implementation evidence, so enabling the project to be evaluated by means of practical results including geospatial mapping, hot spot identification, model performance, explainability, and dashboard visualisation.

Table 2: Project objectives and evaluation criteria.

Objective	Implementation evidence	Evaluation criterion
Map cyber-attack events geographically	IP-based geolocation creates latitude, longitude, country and city; GeoPandas stores point geometry; spatial figures plot mapped events.	Events must be converted into valid spatial coordinates and represented visually.
Identify hotspot and density patterns	DBSCAN clustering, final hotspot score, Getis-Ord Gi* z-score mapping, KDE density mapping and Moran's I.	Hotspot outputs must be produced and interpreted without overstating statistical certainty.
Predict high-risk records	Random Forest, XGBoost, LightGBM, CatBoost and ensemble models are trained and compared.	Model performance must be evaluated through accuracy, F1-score and confusion-matrix evidence.
Explain analytical behaviour	Correlation heatmap, network graph and SHAP summary plot.	The analysis must identify influential features and relationships rather than presenting predictions as a black box.
Communicate findings through dashboard visualisation	Power BI pages present internet and economic context, attack-type and severity analysis, and country deep-dive views.	Dashboard outputs must support visual exploration and communicate project findings clearly to technical and non-technical audiences.

3. Implementation Process

The design was intended as a full-fledged geospatial cybercrime analysis procedure. The procedure used in the project is outlined in Figure 1. Loading and cleaning data starts the process, which then generates ASN, temporal, spatial, and contextual features. Hotspot mapping, pattern analysis, model evaluation, and explainability are then used to examine the enriched dataset. The project requires this sequence

since before any significant spatial or predictive analysis can be done, cyber-attack records must first be geocoded and standardised.

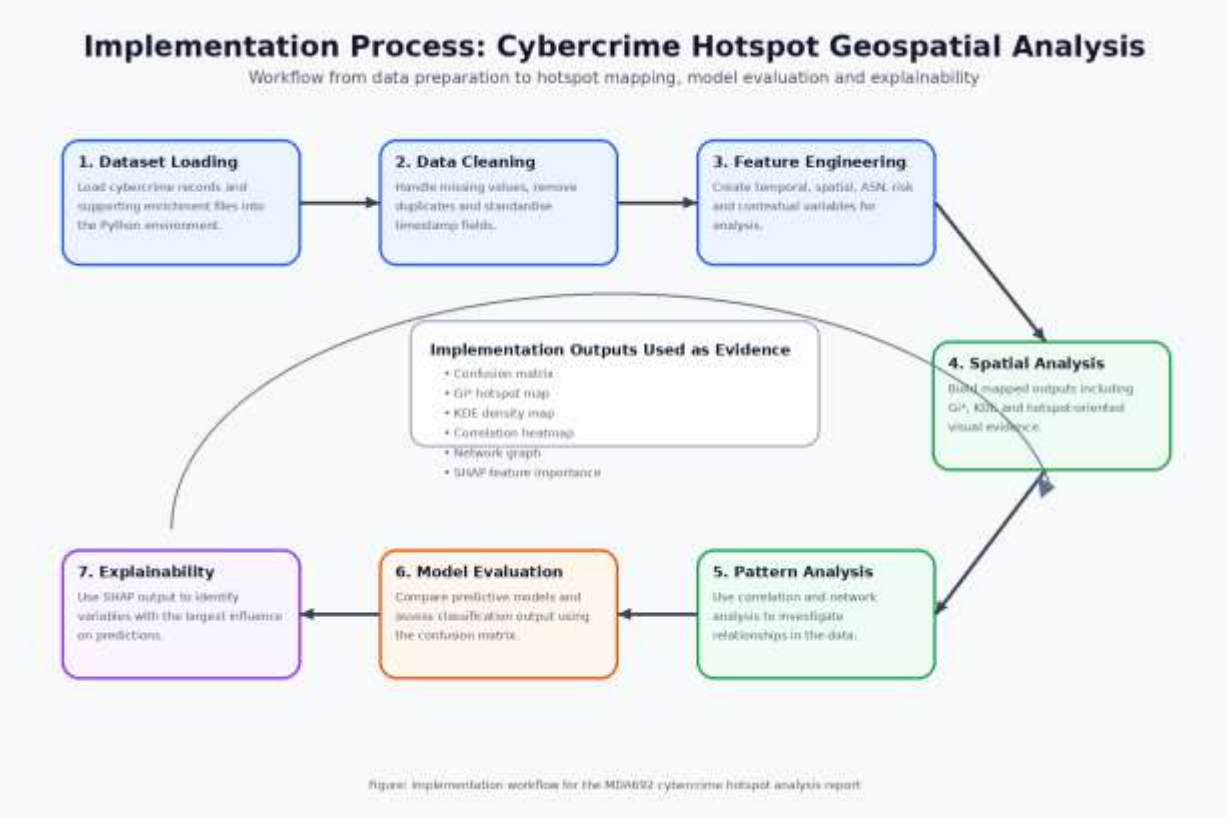


Figure 1: Implementation process for the cybercrime hotspot geospatial analysis workflow.

3.1 Environment and Libraries

The programme makes use of a Python analytics environment with packages for data processing, geospatial analysis, machine learning, visualisation, network analysis, and model explainability. Pandas and NumPy are employed for numerical calculations and tabular data handling. GeoPandas handles spatial data structures; rasterio supports raster sampling; geoip2 uses MaxMind databases to enable IP geolocation. DBSCAN, train-test splitting, random forest, preprocessing, and classification measures all rely on scikit-learn. Other supervised learning techniques come from XGBoost, LightGBM, and CatBoost. Calculations for spatial weights, Moran's I, and Getis-Ord Gi* depend on PySAL. While SHAP helps to understand model behaviour, NetworkX helps to establish graph-based relationships. Folium, matplotlib, seaborn, and static and map-based graphics are supported. For the last presentation layer, Power BI Desktop was utilized as the visual analytics and dashboard reporting tool.

This mix of libraries is perfect as the project calls for more than one approach. It calls for data cleansing, feature creation, geolocation, spatial statistics, prediction, visualisation, and interpretability. As such, the execution captures a realistic data science pipeline rather than a single model experiment.

3.2 Data Loading and Cleaning

Pandas loads the working cyber-attack dataset using latin1 encoding. The cleaning process turns the Timestamp field into datetime format, gets rid of duplicate records and fills in missing values with Unknown. From Timestamp, the programme produces month, day_of_week, and hour. These factors let the analysis include temporal behaviour instead of just depending on location. Standardised to lowercase, text fields including Protocol, Traffic Type, Attack Type, and Severity Level lower the possibility that similar categories are regarded as independent values because of capitalisation variations.

The ipaddress library also converts IP addresses into numbers. Handling incorrect IP values safely entails returning missing values rather than halting the workflow. This is a required step since often cybersecurity data includes errors or malformed values. Cleaning the data before modelling and geolocation helps to lower avoidable mistakes and raises the dependability of following statistical results.

3.3 Geolocation and Contextual Enrichment

Using GeoLite2-City.mmdb, the geolocation stage transforms Source IP Address into Latitude, Longitude, Country, and City. Latitude and longitude are transformed into numerical values and filtered to legitimate geographic ranges following lookup. Coordinates beyond -90 to 90 for latitude or -180 to 180 for longitude are either deleted or deemed incorrect. Invalid coordinates would skew maps and spatial data, hence this quality-control phase is crucial.

The system then adds data on things like how much people use the internet in different countries, how much money they make per person, and how many people live in an area. It also derives local_population_density from a WorldPop raster. At last, GeoLite2-ASN.mmdb includes ASN and ASN_Organization. These elements link infrastructure and local background with cyber occurrences. Consequently, the dataset lends itself to both spatial analysis and forecasting models.

3.4 Feature Engineering and Hotspot Scoring

Severity Level helps to generate the High_Risk target label. While other records get a 0, records labeled as high or critical get a 1. This binary tag is the goal for supervised learning. The implementation also produces country_attack_count, asn_risk_score, and asn_attack_count—all aggregated risk variables. These qualities show the degree of measured activity related to country and network groups.

Attack_per_density, internet_risk_score, economic_risk_score, anomaly_time, and cluster_risk are further designed factors. Together with cluster information, these elements help construct a final_hotspot_score from anomaly score, ASN risk, density-related risk, and internet-related risk. Hotspot analysis shouldn't depend just on raw counts, hence this feature-engineering approach is crucial. Better understood as a mix of unusual behavior, focus, network connection, and environmental exposure is a significant cybercrime hot spot.

3.5 Spatial Analysis, Clustering and Visualisation

DBSCAN clustering is used here on scaled Latitude, Longitude, local_population_density, and ASN risk factors. DBSCAN works well for this project since it can spot thick clusters of data and categorize lone observations as noise. It's helpful when the sites of cyber-attacks are scattered since, unlike K-Means, the analyst is not required to specify the quantity of clusters.

The spatial analysis also features Getis-Ord Gi* and Moran's I. Moran's I examines whether anomaly scores exhibit general spatial autocorrelation. By deriving z-scores for spatial clustering, Getis-Ord Gi* helps to find particular warm and cold locations. Kernel density estimation is then used to show continuous concentration surfaces. These results give different points of view: Moran's I shows general spatial autocorrelation, Gi* shows local hotspot tendencies, and KDE shows how many events happen in an area.

3.6 Machine Learning and Explainability

The supervised learning part analyzes CatBoost, XGBoost, Random Forest, LightGBM, and an ensemble model. Features comprise temporal variables, categorical encodings, spatial coordinates, anomaly scores, contextual factors, and constructed risk scores. Accuracy and F1-score help assess the models; a confusion matrix supports the last prediction result. This comparison is significant since one model outcome is inadequate for a thorough project assessment. Comparing many models helps to show which algorithm is most effective given the existing attributes.

Through correlation analysis and SHAP values, explainability is added. While SHAP reveals which factors most influence model output, the correlation heatmap reveals the relationship between numerical variables. In the context of cybersecurity, this is especially crucial since model predictions should not be regarded as flawless. Analysts have to determine whether the model is driven by weak or misleading variables or if it is affected by relevant features including ASN risk, hotspot score and cluster risk.

3.7 Power BI Dashboard Development

The project's reporting level was created as a Power BI dashboard. Presenting important indicators via graphic pages understandable by both technical and non-technical audience, the dashboard enhances the spatial and machine-learning study. The dashboard is not meant to replace the results of statistical analysis and modelling; instead, it is a way to talk about the things you found and the things that could affect them in a way that is easy to understand and can help people make decisions.

There are three major report pages on the dashboard. The first page, Global Internet and Economic Review, looks at the background of society, the economy, and how well people are connected. It has a scatter plot contrasting GDP per capita with internet usage, a trend chart showing internet usage by country, year, and country slicers, and a world map of internet use by nation. This page helps the project aim of relating cyber-risk interpretation to more general indicators of connection and economy.

The second page, CyberSecurity Attacks Analysis, looks at the data for a cybersecurity event. It shows time-based attack trends, slicers for attack type and severity, a severity-level breakdown, and attack counts by type. DDoS, Malware, and Intrusion are the visible categories; severity is rated as High, Medium, and Low. This page helps exploratory analysis by letting attack patterns be examined without the audience having to look at raw tables.

The third page, Country Deep Dive, gives background information on each country. It offers country filtering, median internet usage, median GDP per capita, median population density, country trends across time, and a ranking of the top nations or areas by internet use. This page helps to understand why cybercrime hotspot analysis should take more into account than only event counts across nations and for comparing connections and demographic background.

4. Testing and Troubleshooting

As part of a series of data-quality, geolocation, spatial-analysis, and model-evaluation checks, testing was done all through the implementation. The goal of testing was not only to verify that code ran

correctly, but also to guarantee that every analytical step generated intelligible and reasonable results. The main testing domains and results are listed in Table 3.

Table 3: Testing and troubleshooting evidence.

Testing area	Method used	Outcome and interpretation
Missing and inconsistent values	Missing values were filled, duplicates were removed and text fields were standardised.	The dataset became more consistent before modelling and visualisation.
Timestamp validity	Timestamp was converted to datetime with invalid conversions handled safely.	Temporal variables could be extracted without stopping the implementation.
IP conversion and geolocation	IP transformation and GeoIP lookup used safe error handling.	Invalid IP addresses or lookup failures did not break the pipeline.
Coordinate validation	Latitude and longitude were converted to numeric values and checked against valid geographic ranges.	Map outputs were protected from impossible coordinates.
Spatial statistics	KNN weights, Moran's I and Getis-Ord Gi* were calculated.	Spatial analysis was implemented; significance was interpreted cautiously where p-values were not displayed.
Model evaluation	Accuracy, F1-score, classification outputs and confusion matrix were reviewed.	Predictive performance could be compared across models.
Visual evidence	Hotspot, density, correlation, network and explainability figures were generated.	The implementation produced interpretable visual outputs for the report.
Dashboard visualisation	Power BI pages were exported to PDF and reviewed for slicers, labels, chart readability and alignment with the implemented variables.	The dashboard pages provide readable summaries of internet/economic context, cyber-attack patterns and country-level indicators.

One major problem was the difference between real analysis and computer software that could be used to help make decisions. Though a full interactive deployment is outside the shown stage, the study generates legitimate static data and model outputs. This difference is addressed by judging the actual analytical prototype on its own data instead of exaggerating it as a live working system.

5. Results and Evaluation

The output produced by the used workflow is assessed here. Classification performance, hotspot mapping, density mapping, feature relationships, network structure and explainable model behaviour are the main points of the assessment.

5.1 Model Performance

Table 4: Model performance comparison.

Model	Accuracy	Class 0 F1-score	Class 1 F1-score	Interpretation
XGBoost	0.7170	0.71	0.72	Moderate predictive performance; better than LightGBM and CatBoost in this run.
Random Forest	0.8524	0.85	0.85	Best single model in the implementation.
LightGBM	0.6761	0.67	0.68	Lower predictive performance on the documented feature set.
CatBoost	0.6709	0.67	0.68	Similar to LightGBM and below XGBoost.
Ensemble	0.7347	0.75	0.72	Improves over XGBoost, LightGBM and CatBoost, but not over Random Forest.

With 0.8524, Random Forest had the highest accuracy; it also had balanced F1-scores of around 0.85 across both classes. This implies that tree-based ensemble learning fits the feature set quite well. Though it improved upon some single boosting techniques, the ensemble model achieved 0.7347 accuracy and did not surpass Random Forest. Therefore, Random Forest is the most efficient single model for this feature configuration and dataset given the recorded implementation.

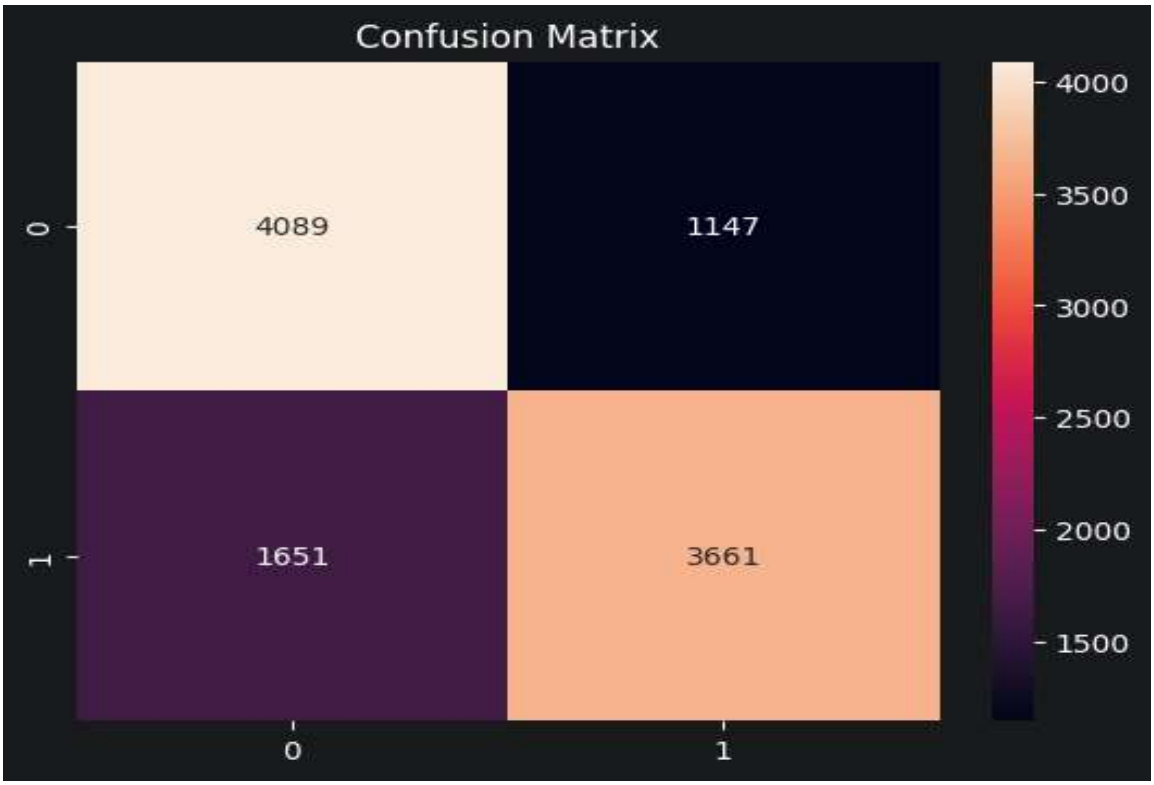


Figure 2: Confusion matrix for the final prediction output.

The confusion matrix shows 4,089 true negatives, 3,661 true positives, 1,147 false positives and 1,651 false negatives. Therefore, the model accurately finds a large number of low-risk and high-risk records but misses some high-risk observations. In the setting of cybersecurity, false negatives are significant since they identify high-risk records that the model deems low-risk. If the model is used for operational prioritisation, the false-negative count so suggests that future research should concentrate on enhancing recall for the high-risk group.

5.2 Hotspot and Density Mapping

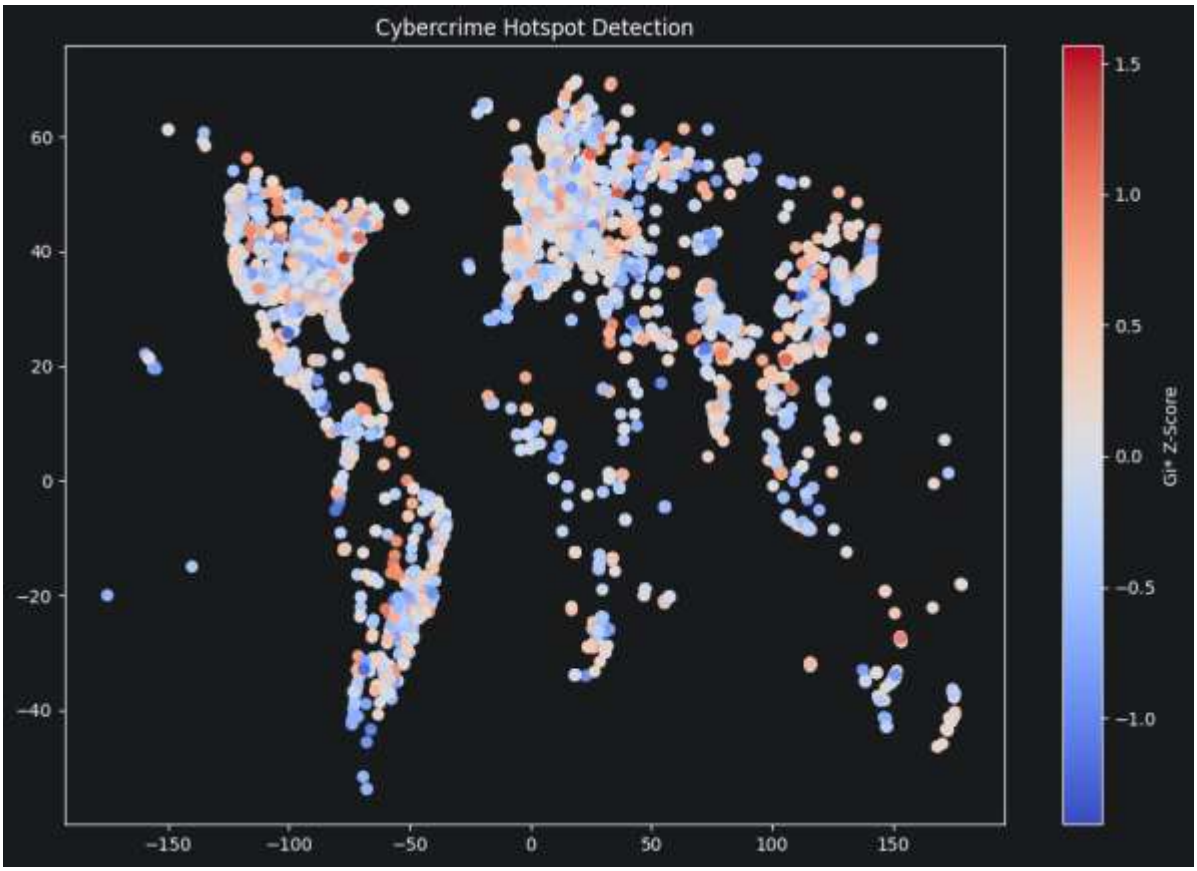


Figure 3: Getis-Ord Gi cybercrime hotspot detection map.*

Getis-Ord Gi* hotspot identification output is shown in Figure 3. The map colours events by Gi* z-score and positions them all around the world. Positive z-scores show locations where lower values are locally concentrated; negative values point to clustering of higher values. The charted data reveals that records are dispersed over Oceania, Asia, Africa, Europe, South America, and North America. Although local high-score points show up in several places, the map should be viewed as a hotspot-screening output rather than a final claim of named real-world attack sites since the map is not labeled by city or nation and IP geolocation is approximate.

For anomaly scores, Moran's I implementation report shows 0.00339. This value shows extremely faint positive spatial autocorrelation. Practically speaking, this implies that under this execution the anomaly-score layer exhibits weak general worldwide clustering. This conclusion demonstrates that local hotspot behaviour and global autocorrelation are distinct ideas rather than discrediting the hotspot map. Local Gi* clustering can still find local patterns even if the value of Moran's I overall is low.

The Getis-Ord Gi analysis gave a look at local hotspot significance. Out of 36,200 records analysed which is 91.31% of the total no significant statistics were found. However 1,178 records or 2.97% showed hotspots at a 90% confidence level. Also 888 records or 2.24% were hotspots at a 95% confidence level. 135 Records or 0.34% were hotspots at a 99% confidence level. In total 2,201 records or 5.55% were hotspot observations. The Getis-Ord Gi analysis results help to understand that cybercrime risk is concentrated in some areas. These areas are not spread out evenly across the dataset. The Getis-Ord Gi analysis finding also matches the Moran's I result. The Moran's I result shows that global spatial autocorrelation is limited. Local hotspot patterns from the Getis-Ord Gi analysis are still there. The Getis-Ord Gi analysis provides precise information. It shows that hotspot observations from the Getis-Ord Gi analysis are concentrated. The concentration is, in selected areas.

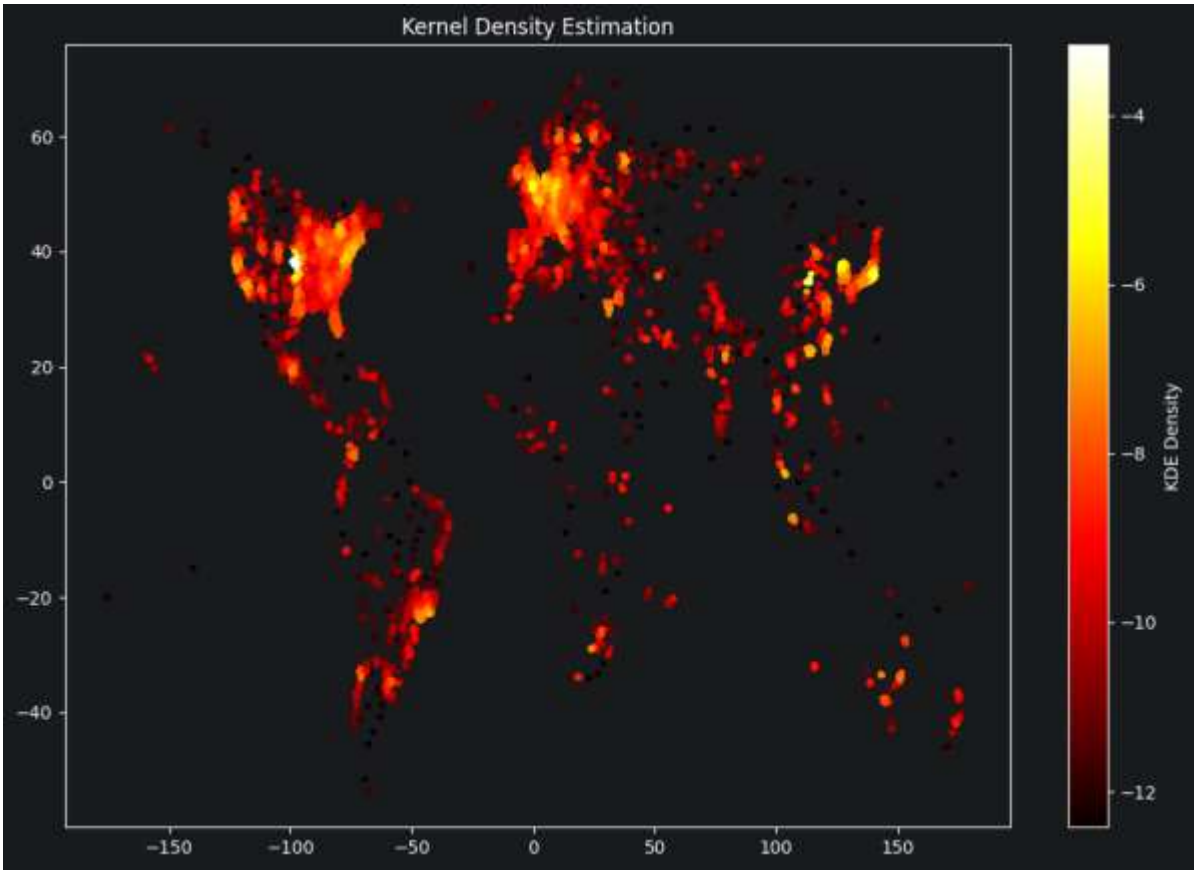


Figure 4: Kernel density estimation of mapped cyber-attack events.

The kernel density estimate output is seen in Figure 4. Lighter areas show more density, whereas darker areas show less concentration. The KDE map is useful because it turns individual event points into a continuous surface that shows how dense something is. This helps one to spot general concentration patterns more readily than a basic scatter plot. The most obvious concentrations may be found in areas

of North America and Europe; other inhabited locations also show concentrations. The map also emphasizes the necessity to analyze density trends in conjunction with data coverage and geographic limits since it also shows few observations in less densely represented regions.

Folium interactive heat map output

A Folium heat map was also made from the cyber-attack records that had location information. The notebook figures out the center of the map using the latitude and longitude values. It then starts a Folium map that shows the world and adds the cleaned up location points to create a heat map. This is really useful because it lets you look at the areas with a lot of cyber-attacks in detail by moving the map around and zooming in. The other map in Figure 4 is good for a summary. This one is better, for exploring the locations of the cyber-attacks.

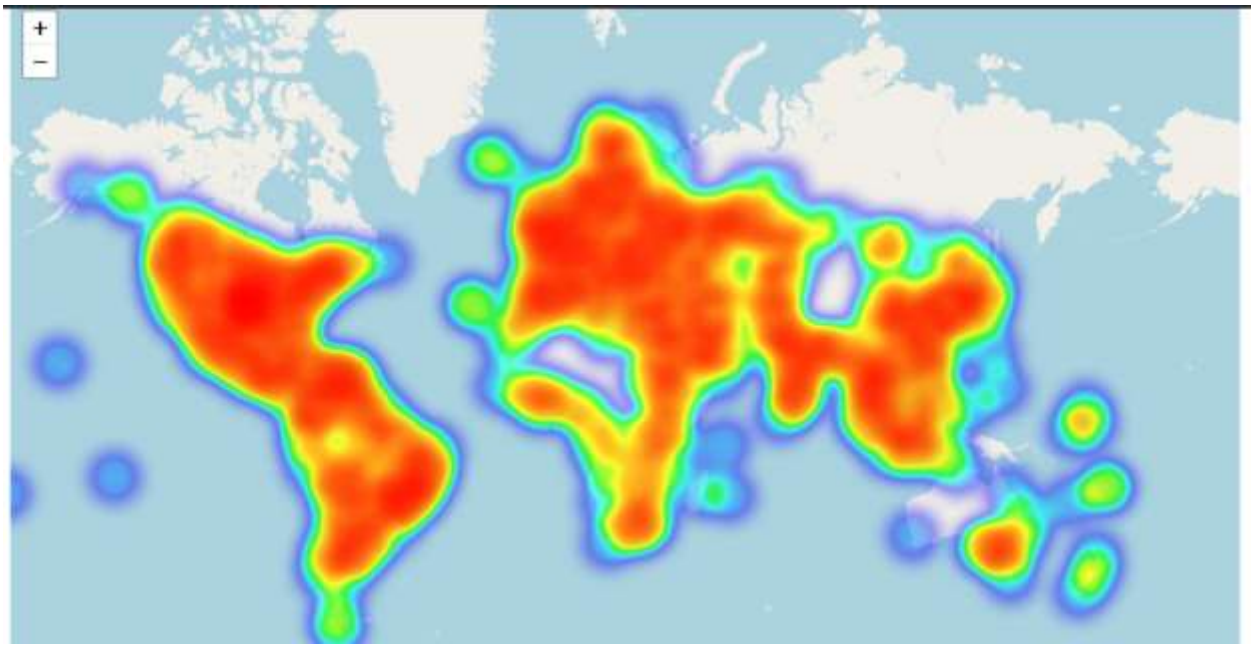


Figure 4a: Folium heat map showing the global concentration of geolocated cyber-attack events.

The heat map shows that events are not spread out around the world. There are events in certain areas like North America, Europe and some parts of Asia, where the same IP addresses are located. We should not think that these areas are where the offenders are exactly. IP addresses can be linked to things like internet service providers, VPNs or cloud hosting areas. So the heat map helps us see patterns. We need to look at other results like Gi*, KDE, Morans I and model evaluation to get a better understanding. The heat map is one tool to help us explore the data. It confirms that event density varies across the world. We see concentrations of events in specific regions. These regions include North America, Europe and

parts of Asia. The IP geolocation can be linked to routing infrastructure. It can also be linked to VPN endpoints, proxy services or cloud hosting regions. Therefore we should consider the Folium output along, with results. The other results include Gi*, KDE, Morans I and model evaluation results..

5.3 Feature Relationship Analysis



Figure 5: Correlation heatmap of spatial, contextual, temporal and risk features.

The correlation heatmap can be seen in Figure 5. Engineered risk factors link to the best relationships. Internet_risk_score and attack_per_density are strongly correlated with final_hotspot_score; it also has a strong association with economic_risk_score. This makes sense given that the hotspot score is calculated from a number of these risk variables. The heatmap further reveals that, in this implementation, High_Risk has quite poor links with temporal factors including month, day_of_week,

and hour. This implies that, given the present data and features, spatial, ASN, and created-risk elements are more helpful than just basic calendar-time indicators.

The heatmap even points to any potential multicollinearity. For instance, internet_risk_score and economic_risk_score have a strong correlation, which suggests that some designed contextual risk factors could overlap. For tree-based models, this isn't always a concern, but it's something to keep in mind when simplifying models going forward. Keeping both variables could obscure interpretation even if predictive performance is good if two variables hold essentially the same information.

5.4 Network Structure

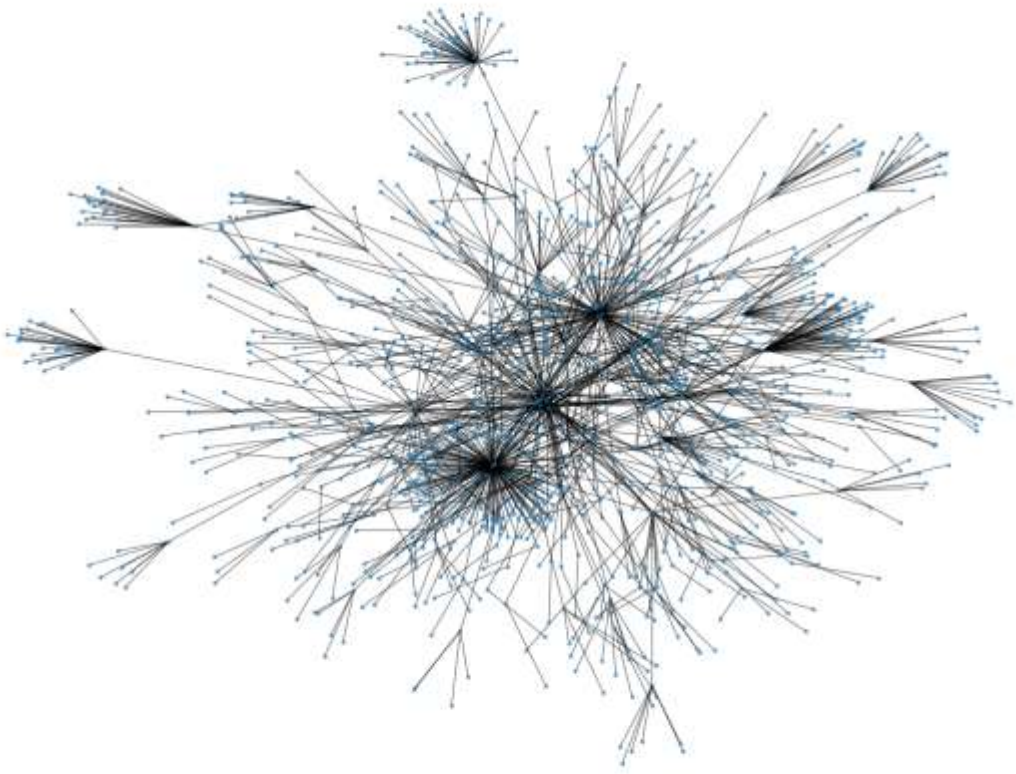


Figure 6: Network graph showing relationships among selected cyber-event entities.

Figure 6 shows a network graph. The visual layout indicates a thick central group surrounded by several smaller peripheral clusters. This implies that while some companies in the data serve as nodes linking several observations, others seem as smaller groups or solitary branches. Such a graph is helpful in cybercrime research since it exposes whether particular IP addresses, ASNs, nations, or attack types

often relate to several records. The visual density around key nodes shows that network relationships can offer more information than conventional table summaries would.

The graph is not accepted as actual evidence of criminal organization. Rather, it serves as initial proof of recurring patterns in the data. This difference is crucial since shared infrastructure, common service providers, routing patterns, scanning behaviour, or repeated attack signatures might all cause network links in cyber logs to emerge. As a research tool for ranking more analysis, the graph is so most valuable.

5.5 Explainability with SHAP

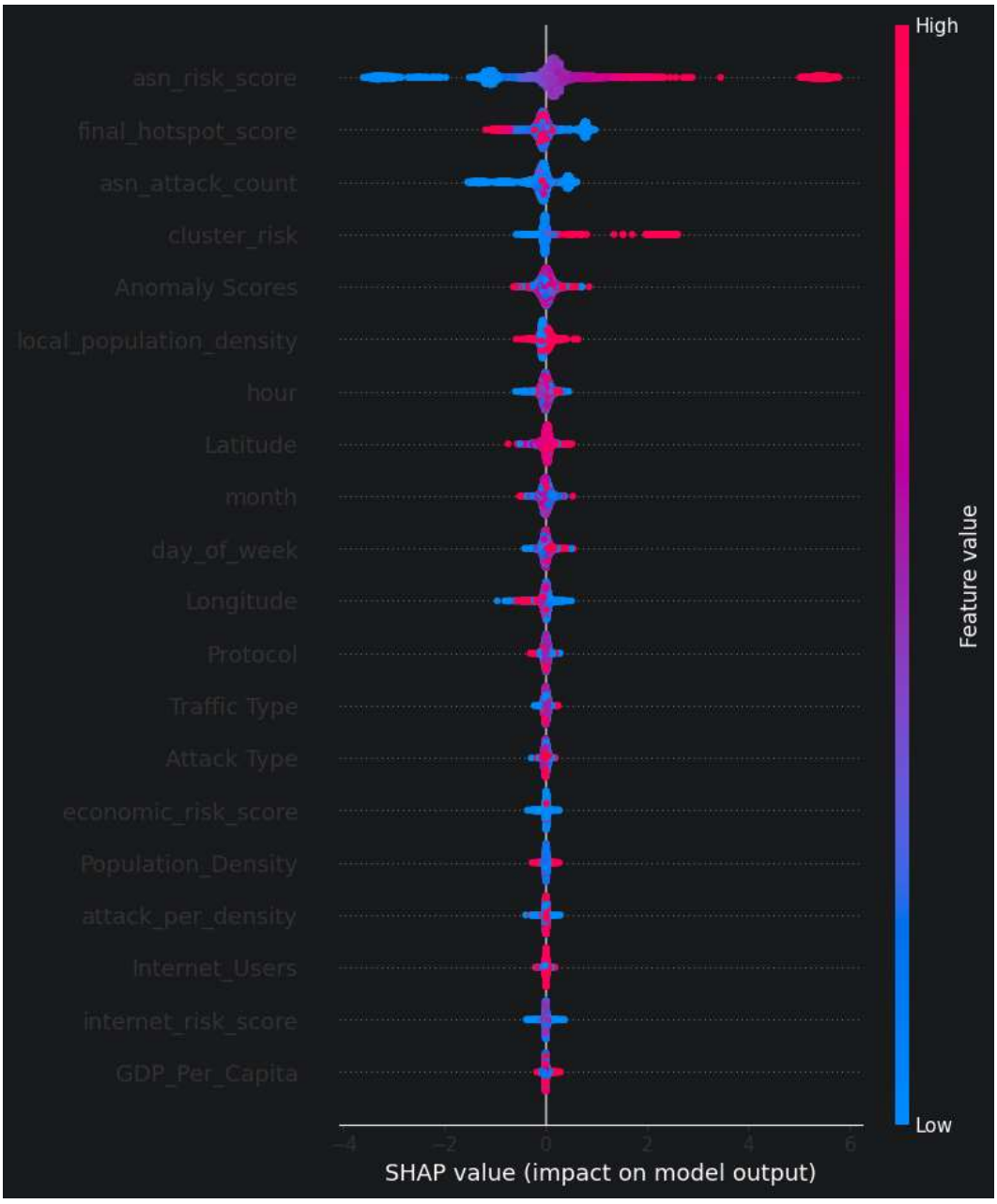


Figure 7: SHAP summary plot showing feature influence on model output.

The SHAP summary plot is shown in Figure 7. The most important factor is asn_risk_score, then comes final_hotspot_score, asn_attack_count, cluster_risk, and Anomaly Scores. This ranking matters since it demonstrates that the model is heavily impacted by network-level risk, engineered hotspot scoring, and clustering data rather than just on raw geographic coordinates. This justifies the design choice of adding ASN and hotspot-related features to the dataset.

The SHAP result further suggests that several factors, including GDP_Per_Capita, Internet_Users, and particular temporal variables, have little bearing on prediction. This does not imply in all cybercrime studies these factors are insignificant. It just indicates that in the recorded implementation and feature set, they contributed less to the model's ultimate predictions. This is a helpful result since it enables one to differentiate between theoretically significant traits and traits actually important in the model used.

5.6 Power BI Dashboard Results and Evaluation

There is also a Power BI visual element in the project. The implementation output includes an evaluation of the dashboard pages in this section. The dashboard helps non-technical viewers, business stakeholders, and managers understand the analytical dataset and contextual indicators by presenting them as visual summaries that promote communication.

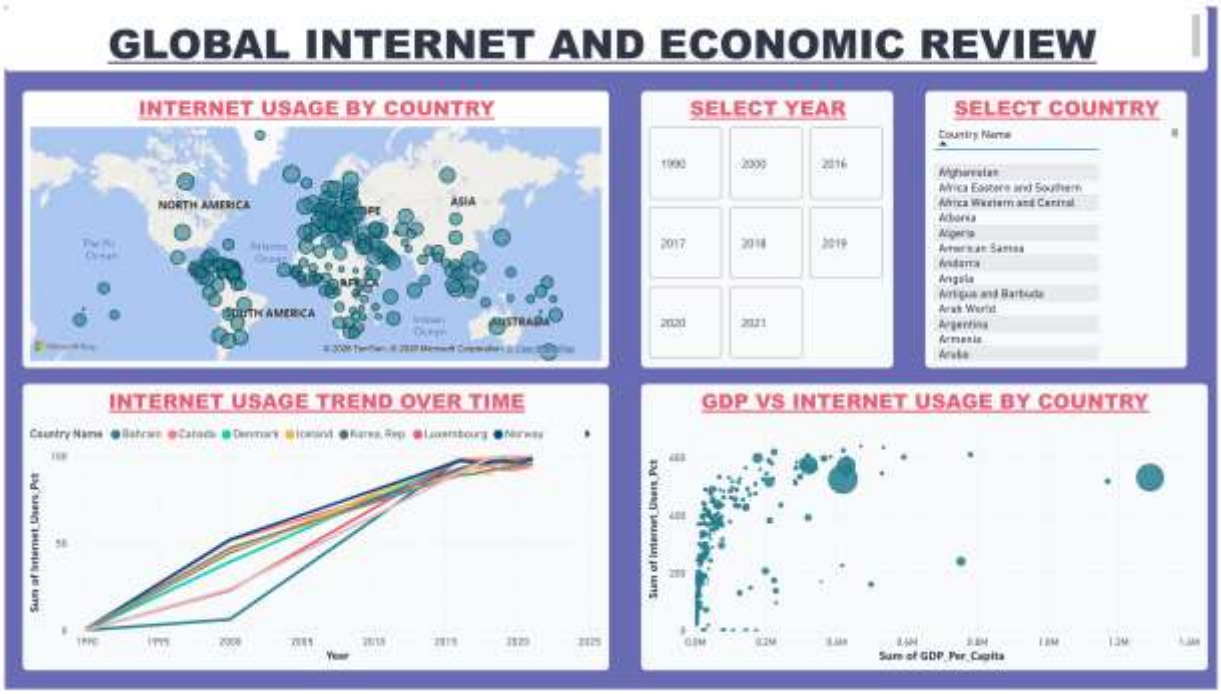


Figure 8: Global Internet and Economic Review dashboard page.

Page Global Internet and Economic Review is seen in Figure 8. The dashboard combines a geographical map, year slicer, country slicer, internet-usage trend chart, and GDP-versus-internet-usage scatter plot. The map gives a quick worldwide summary of internet usage by nation, whereas the trend chart shows that chosen nations typically saw significant internet usage increase from 1990 through the 2010s and early 2020s. The scatter plot shows that while there is a correlation between GDP per capita and internet use, it is not exactly linear; some countries and regional aggregates have high rates of internet usage yet do not sit precisely on the GDP scale. The page is significant for this project since cybercrime risk has to be understood in connection with digital exposure, connection, and economic environment rather than just the quantity of cyber events.

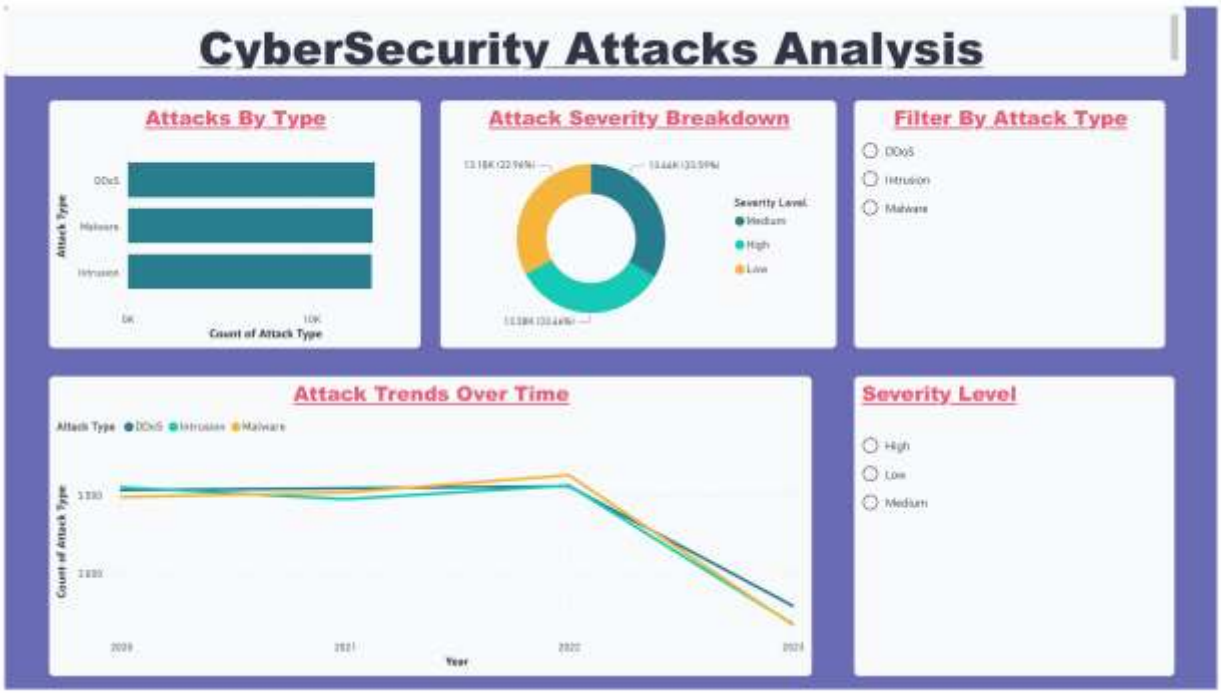


Figure 9: CyberSecurity Attacks Analysis dashboard page.

The CyberSecurity Attacks Analysis website is shown in Figure 9. The bar graph contrasts attacks by kind and displays the three adopted groups: DDoS, Malware and Intrusion. The donut chart shows how many records there are in each severity level. Medium, High, and Low severity records each make up about one-third of the dataset. With around 13.44K Medium records, 13.38K High records, and 13.18K Low records, the visible values show a balanced severity distribution instead of a dataset mostly made up of one class. Over 2020 to 2023, the time-trend graph shows DDoS, Intrusion, and Malware. From 2020 to 2022, the trend is fairly constant; in 2023 there is a clear drop. The page is helpful for concentrated

review because users can narrow down particular attack or severity levels using the slicers for attack type and severity.

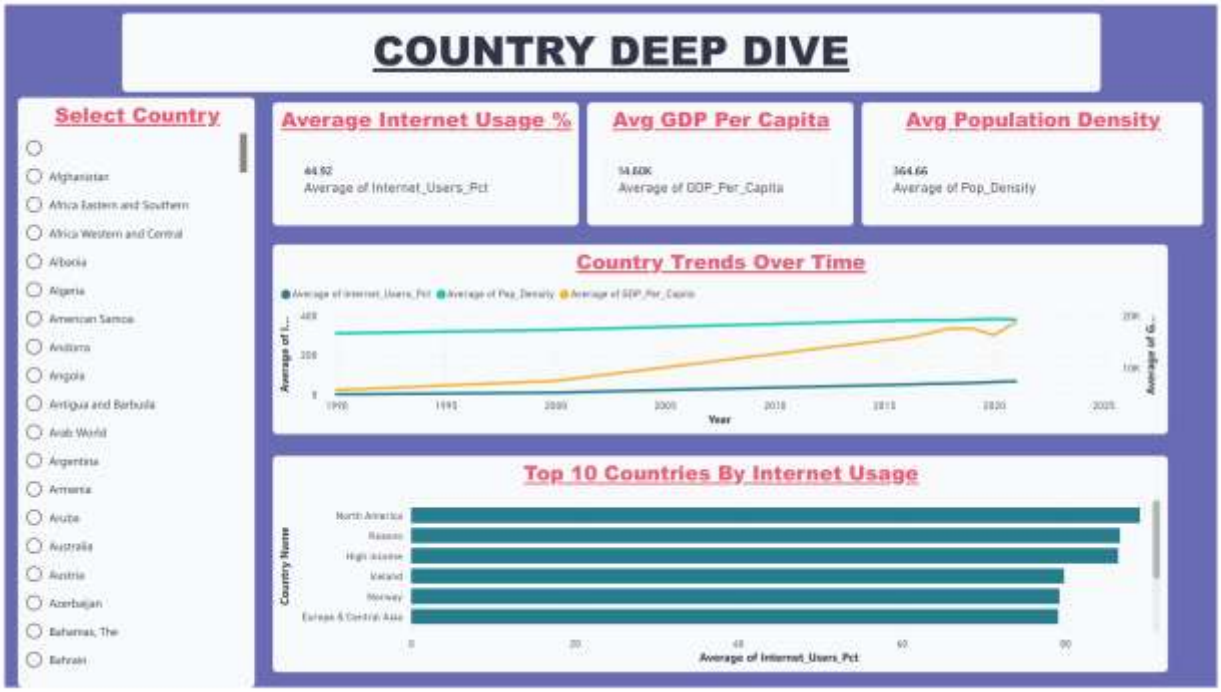


Figure 10: Country Deep Dive dashboard page.

The Country Deep Dive page is shown in Figure 10. Using cards, a country trend chart, and a top-10 internet-usage rating, this page compiles country-level contextual information. For the chosen perspective, the dashboard shows an average internet-usage value of 44.92%, an average GDP per capita of 14.60K, and an average population density of 364.66. The country trend line displays population density, GDP per capita, and internet usage over time; the ranking chart spotlights nations and areas with great average internet usage, including North America, Kosovo, High income, Iceland, Norway, and Europe & Central Asia in the visual ranking. By showing how country-level indicators range across the dataset and by facilitating the interpretation of the contextual enrichment phase, this page helps the project.

The dashboard essentially enhances the initiative with a communication and exploration layer. The machine-learning and spatial-statistical findings offer analytical depth, whereas the Power BI pages assist to show the larger picture: internet connection and economic development give background, attack type and severity define cybersecurity behaviour, and country-level indicators help to explain why

geographical interpretation calls for care. Therefore, the dashboard makes the project more user-friendly without overemphasizing it as a live operational monitoring tool.

5.7 Overall Evaluation Against Objectives

Table 5: Evaluation against project objectives.

Objective	Status	Evidence	Evaluation
Geographic mapping	Completed	GeoIP coordinates, valid latitude-longitude filtering, Gi* map and KDE map.	The project successfully converts event records into spatial outputs.
Hotspot identification	Completed with cautious interpretation	DBSCAN clustering, Gi* z-scores, KDE and Moran's I.	Local hotspot screening is achieved, but weak Moran's I and geolocation uncertainty limit overstatement.
High-risk prediction	Completed	Five models compared, with Random Forest achieving 0.8524 accuracy.	Predictive modelling is a strong component of the implementation.
Explainability	Completed	Correlation heatmap, network graph and SHAP plot.	The model and risk variables are interpreted rather than presented as unexplained outputs.
Dashboard communication	Completed	Three Power BI dashboard pages: Global Internet and Economic Review, CyberSecurity Attacks Analysis, and Country Deep Dive.	The dashboard strengthens stakeholder communication by connecting raw analysis with interactive business-intelligence style views.

6. Comparative Analysis with Literature Review

Firstly, our literature review indicated that cyber-crime and cyber-enabled crime have a variety of dimensions such as spatial, temporal, socio-economic and infrastructural perspectives. The implementation we have done is in-line with that foundation by showing how cyber-attacks can be analyzed by enriching cyber attack logs with geographic and network features. In addition to that, there are some practical issues in terms of geolocation uncertainty, feature overlap and the weakness of global spatial autocorrelation as well as the prototype vs. dashboard discussion.

As Wang and Li [4] proved the power of spatiotemporal crime models with their use of crime data sets enriched by geography analysis, it can be assumed that time and geographic location of crimes can show patterns in data which cannot be seen from the non-spatial perspective. Thus, our present implementation does follow this logic by using features like hour, day_of_week and month and geolocations of cyber-attack logs. Still, the main difference is the use of IP based cyber data set versus the official police crime data set.

Malicious URLs have been studied using spatial analysis in research carried out by Amin et al. [5], which demonstrates how cyber indicators could be transformed into mappable risk indicators. The current study is in line with this research in the sense that it also uses coordinates derived from source IPs and generates Gi* and KDE maps. The main difference is that the current implementation has also included risk from ASNs, local population density, country context indicators, and machine learning predictions. In that regard, this implementation enhances the mapping aspect into a more comprehensive risk modelling process.

According to Hall et al. [6] and Chen et al. [7], cybercrime has links with economic geography, infrastructure, connectivity, and global inequalities in general. The current implementation considers aspects of internet use, GDP per capita, population density, and local population density in response to these research findings. The results indicate that not all contextual variables had an equal importance in the current model, but they are important for testing the hypotheses behind using such context in a model rather than only using coordinates.

This is addressed by Chen et al. [8] and Zhuo et al. [9], who emphasise the roles of regional factors, temporal dynamics, and spatial clusters in cyber-fraud and cybercrime. This proposed methodology is aligned with this trend because it generates hotspot scores, uses DBSCAN, and considers local Gi* patterns. However, Moran's I is extremely low in the demonstrated output, suggesting that there is no strong overall spatial autocorrelation pattern in cyber-attack risks in this dataset. This observation indicates that a cautious approach should be taken in interpreting results from cybercrime data.

Table 6: Comparison between literature findings and the implemented project.

Literature finding from MDA691	Reference	Comparison with current implementation	Implication
Crime and cyber-enabled crime can vary spatially and temporally.	Wang and Li [4]; Zhuo et al. [9]	The implementation derives temporal variables and maps attack records geographically.	Cybercrime should be analysed through time and place, not only through total counts.
Cyber indicators can be converted into spatial hotspot evidence.	Amin et al. [5]	Source IP addresses are geolocated and analysed through Gi* and KDE maps.	Spatial visualisation is a valid method for hotspot screening.
Cybercrime is connected to economic, connectivity and infrastructural context.	Hall et al. [6]; Chen et al. [7]	Country-level internet, GDP and population indicators are merged into the dataset.	Contextual enrichment strengthens interpretation beyond raw cyber logs.
Regional drivers and clustering methods are useful for cyber-fraud analysis.	Chen et al. [8]	DBSCAN, final hotspot score, Gi* and Moran's I are implemented.	Local hotspot methods are useful, but global spatial autocorrelation should not be overstated.

7. Project Contribution and Professional Considerations

The primary value provided by this project is the implementation of a combination of several levels of analysis into a single geospatial cybercrime analysis pipeline. The current work goes beyond simple visualization or modelling. Cybersecurity events have been enriched with geolocation by IP address and ASN, combined with socioeconomic data by countries, sampled with a raster population method, assigned labels, clustered, scored, supervised learning applied, and network analysis carried out with explainability applied.

From the point of view of the professional practice standards for data analytics capstone projects, the project meets all expectations from this perspective. First, it includes application of technical knowledge about data cleaning, feature engineering, geospatial and machine learning methods. Second, it uses scientific research on the topic as the foundation for implementation decisions. Third, it acknowledges limitations of results. Finally, it delivers visualizations with clear explanatory potential.

Interpretation of results obtained using IP geolocation methods is especially important when discussing ethical implications of such data analysis projects. IP geolocation cannot serve as grounds for accusations of anyone or any organization since it could potentially contain inaccuracies. The result

should be treated as analytical indicators rather than facts that could be used without proper consideration and validation.

The finished implementation also provides a distinct direction for further development. The existing analysis and Power BI dashboard can be further developed into a production system using validated real-time feeds, storing geospatial data into PostGIS, releasing a secure interactive dashboard and doing user testing with cybersecurity subject matter experts. These developments would be seen as next-stage activities, not part of Assignment 2's implementation completion.

8. Conclusion, Limitations and Future Work

8.1 Conclusion

Assignment 2 implementation is a great success as it has effectively achieved the goal of creating a functional geospatial cybercrime analysis tool. The implementation involves the following tasks: loading event data, data cleaning and standardization, adding geospatial attributes to IP addresses, merging socioeconomic and population context data, engineering risk variables, computing hotspot-specific scores, performing clustering and spatial analyses, training different classifiers, and generating analytical graphics. As for classification, Random Forest is the best individual classifier, having the highest accuracy of 0.8524. Additionally, according to SHAP analysis, ASN risk score, final hotspot score, ASN attack count, cluster risk and anomaly scores are the key predictors for Random Forest output.

The generated maps show the possibility of transforming cyber-event data into hotspot and density maps. Thus, Gi* map gives the opportunity for hotspot screening and KDE map visualizes events' concentrations. Further on, one could observe variable correlations and relations using the heatmap and network graph. In addition, Moran's I test result is insignificant and maps have no labels by their regions. Therefore, the conclusion which is to be drawn is that this assignment provided us with effective risk intelligence, yet not solid evidence of real-life offenders' presence at particular regions.

8.2 Limitations

• Geolocation based on IP address is imprecise and may be skewed by factors such as use of a virtual private network (VPN), proxy servers, cloud computing servers, routing behavior, etc.

• The Gi* and KDE maps offer valuable spatial filtering functionality but are unable to pinpoint cities or administrative regions.

• The socioeconomic indicators of countries provide contextual information and cannot be regarded as causes at a local level.

8.3 Future Work

Future research directions could be to enhance validation, deployment and explainability. This would mean incorporating statistical testing with values of p-value for Moran's I and Getis-Ord Gi* results. Spatial results should be labelled according to country, city or administration area for better interpretation of hotspots. In addition, future work will also be needed to incorporate extra methods of validation like cross-validation, classification imbalanced and risk threshold tuning. False negatives are key aspects in cybersecurity, thus future research could be focused on optimising the model for higher recall of high-risk areas.

Secondly, future work could be in the aspect of implementation. The prepared spatial dataset could be saved in PostGIS while the existing Power BI dashboard could be developed to become a published dashboard or a web-based GIS platform. Future work should also include new filters according to the type of attack, time, severity, country, ASN and model risk. Finally, stakeholders' feedbacks according to security analysts or any other supervisor could be collected to validate if the visualised results are comprehensible enough.

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